

Synthetic-data characterization and per-category distribution — Indonesian ASR corpus (v7)

Article-supplement type: *Data in Brief* synthetic-data disclosure (Elsevier sciencedirect-elsevier-format, ISSN 2352-3409). **Generated:** 2026-05-30. **Source:** metadata/dataset_metadata_clean.csv (102,544 rows), dataset_version = v7_natural_synth; reproducible via build_synthetic_report.py (no audio-tree traversal). Machine-readable summary in synthetic_data_stats.json; per-category table in synthetic_per_category.csv.

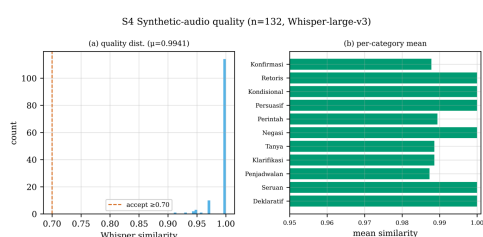
This supplement gives the **full, per-category distribution** of the synthetic gap-fill files used in the corpus that feeds the nine-model fair-comparison benchmark. It complements §1.6 / §2.3 of the main dataset-statistics report and Figure F10 (corpus-level disclosure).

1. Overview

Property	Value
Synthetic files	132 of 102,544 (0.129 % of corpus)
Synthetic duration	0.176 h (632.5 s)
Generation engine	Microsoft Edge-TTS Neural (microsoft_edge_tts_neural)
Voices (gender-matched)	id-ID-ArdiNeural (male, 73) / id-ID-GadisNeural (female, 59)
Speaker-cloned	No — TTS voices only, disclosed (not impersonating speakers)
Synthesis rounds	v7_initial (124) + v7_residual_fix (8)
Speakers with ≥ 1 synthetic	18 / 20
Zero-synthetic speakers	Baron, Robi (both in the test split → clean evaluation)

Synthetic files replace dropped or unrecoverable speaker takes. Each was transcribed with Whisper-large-v3 and required text-similarity ≥ 0.70 to the target sentence before acceptance.

2. Quality verification (Figure S4)

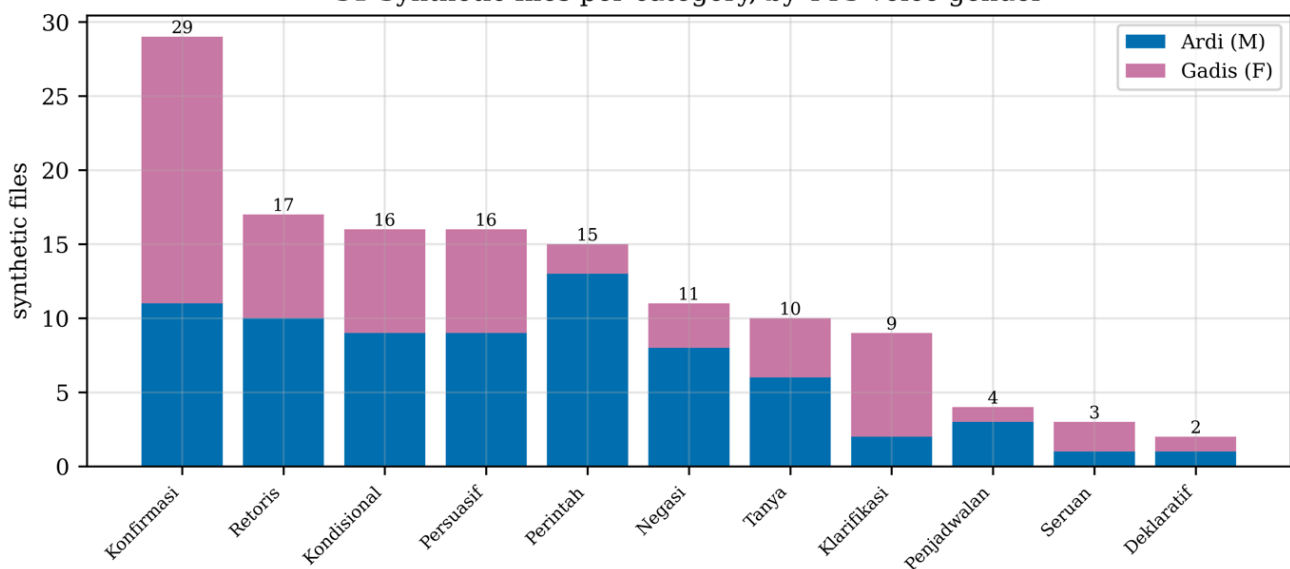


Metric	Value
Similarity metric	Whisper-large-v3 text-similarity to target (0–1)
Acceptance threshold	≥ 0.70
Mean / median	0.9941 / 1.0000
Min / max	0.9101 / 1.0000
Pass ≥ 0.90	132 / 132 (100 %)
Pass ≥ 0.95	125 / 132 (94.7 %)
Exactly 1.00	114 / 132 (86.4 %)

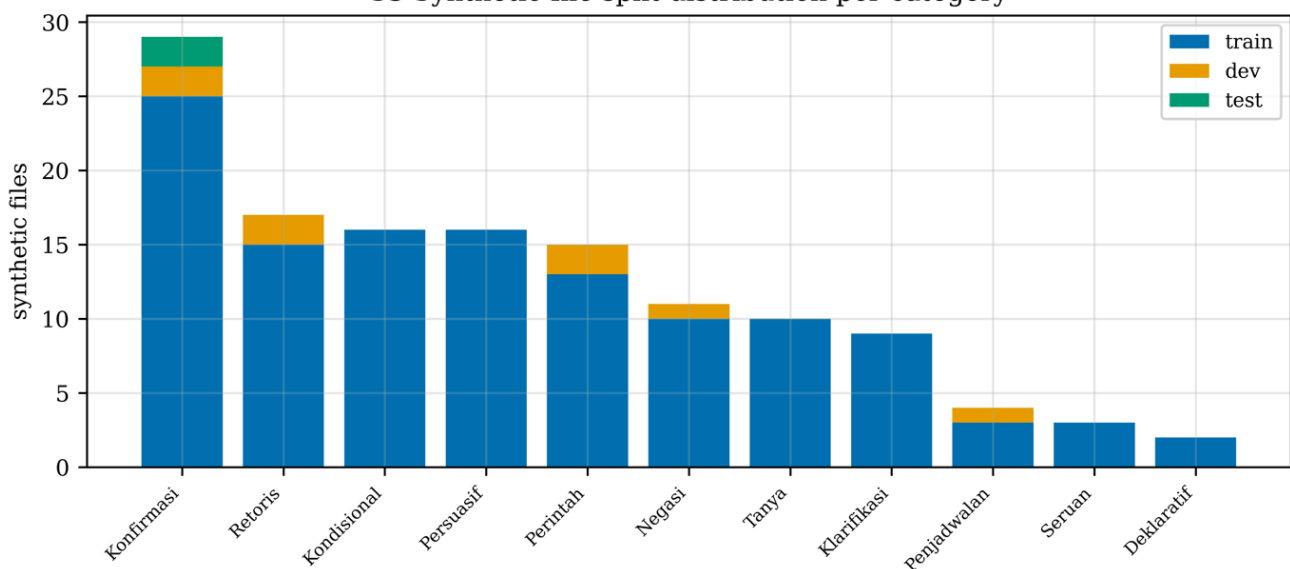
All 132 files clear the acceptance threshold; the lowest-quality file (0.9101, in *Kalimat_Perintah*) still exceeds 0.90.

3. Per-category distribution (Table S1, Figures S1–S3)

S1 Synthetic files per category, by TTS voice gender



S3 Synthetic-file split distribution per category

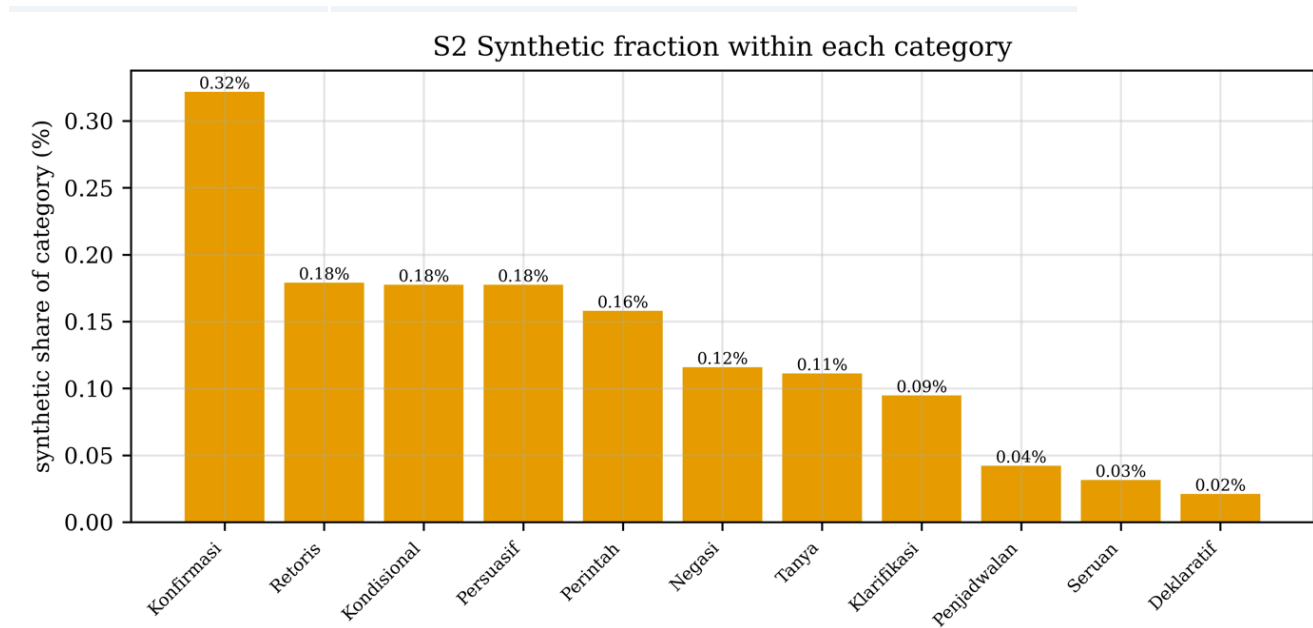


The 132 synthetic files are not uniformly distributed: *Kalimat_Konfirmasi* (confirmation) holds the most (29 files, 21.97 % of all synthetic), while *Kalimat_Deklaratif* holds the fewest (2). Even the most-affected category is only **0.32 % synthetic** of its own 9,017 files.

Table S1. Per-category synthetic-file distribution. %cat = synthetic share within that category; %all = share of all 132 synthetic files; M/F = TTS voice gender (Ardi/Gadis); tr/dv/te = train/dev/test counts; init/fix = v7_initial / v7_residual_fix rounds.

Category	n	%cat	%all	M/F	tr/dv/te	init/fix	$\bar{q}$	q-min
Kalimat_Konfirmasi	29	0.322	21.97	11/18	25/2/2	28/1	0.9878	0.9553
Kalimat_Retoris	17	0.179	12.88	10/7	15/2/0	17/0	1.0000	1.0000
Kalimat_Kondisional	16	0.178	12.12	9/7	16/0/0	14/2	1.0000	1.0000
Kalimat_Persuasif	16	0.178	12.12	9/7	16/0/0	15/1	1.0000	1.0000
Kalimat_Perintah	15	0.158	11.36	13/2	13/2/0	12/3	0.9894	0.9101
Kalimat_Negasi	11	0.116	8.33	8/3	10/1/0	10/1	1.0000	1.0000
Kalimat_Tanya	10	0.111	7.58	6/4	10/0/0	10/0	0.9886	0.9430
Kalimat_Klarifikasi	9	0.095	6.82	2/7	9/0/0	9/0	0.9886	0.9489
Kalimat_Penjadwalan	4	0.042	3.03	3/1	3/1/0	4/0	0.9873	0.9493
Kalimat_Seruan	3	0.032	2.27	1/2	3/0/0	3/0	1.0000	1.0000
Kalimat_Deklaratif	2	0.021	1.52	1/1	2/0/0	2/0	1.0000	1.0000

→ Figure S1 (per-category by voice gender), Figure S2 (synthetic fraction



within each category), Figure S3 (per-category split distribution).

4. Per-split distribution

Split	Synthetic	Split total	% of split
train	122	71,792	0.170 %
dev	8	15,376	0.052 %

test	2	15,376	0.013 %
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Only 2 synthetic files (both in *Kalimat_Konfirmasi*) reach the test split, so the held-out evaluation that scores the nine models is effectively on real speech.

5. Per-speaker distribution

Synthetic files span 18 of 20 speakers (max 17 = Afgan; min 2 = Amri, Joni). Test speakers **Baron** and **Robi** carry zero synthetic files; the third test speaker (Joni) carries 2. Full counts in `synthetic_data_stats.json:by_speaker`.

6. Reproducibility

```
python3 synthetic_data_report/build_synthetic_report.py
# → synthetic_data_stats.json, synthetic_per_category.csv,
#   SYNTHETIC_DATA_REPORT.txt, figures/S1..S4.{pdf,png}, this .md, .pdf
```

Outputs in five formats: **PDF** (this report), **PNG + PDF** (figures S1–S4, 600 dpi / vector, Okabe-Ito palette, `pdf.fonttype=42`), **JSON** (`synthetic_data_stats.json`), **TXT** (`SYNTHETIC_DATA_REPORT.txt`), and **MD** (this file). All numbers trace to `metadata/dataset_metadata_clean.csv`.

Figure index

Fig	Content	File
S1	Synthetic files per category, by TTS voice gender	figures/S1_synth_per_category_voice.pdf
S2	Synthetic fraction within each category	figures/S2_synth_fraction_per_category.pdf
S3	Synthetic-file split distribution per category	figures/S3_synth_split_per_category.pdf
S4	Synthetic-audio Whisper-similarity quality	figures/S4_synth_quality.pdf